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1. Supplementary Methods

1.1. Method S1:Data Source and Definitions for GBD

Our data on the burden of ischemic heart disease (IHD) attributable to kidney dysfunction were from the Global Burden of Disease (GBD) study managed by Institute for Health Metrics and Evaluation (IHME). We accessed the data of the GBD study through visualization platform of Global Health Data Exchange (GHDx) (https://vizhub. healthdata. org/gbd-results/) [1]. Kidney dysfunction was defined by clinical markers, estimated glomerular filtration rate (eGFR) was less than 60 ml/min/1. 73m² or urinary albumin-to-creatinine ratio (ACR) >30 mg/g [2]. IHD, or coronary artery disease (CAD), or atherosclerotic cardiovascular disease (ACD), refers to different diseases such as acute myocardial infarction or chronic ischemic heart disease, which includes angina pectoris and asymptomatic ischemia. GBD assesses the effect of kidney dysfunction on morbidity and mortality and how kidney dysfunction affects cardiovascular morbidity and mortality [2].

1.2. Method S2:Geographical and Demographic Stratification

Analysis stratified data in terms of multiple geographic levels, namely global estimates, epidemiological regions defined by the GBD project, and 204 countries and territories. To examine the association with development level, countries were stratified into 5 quintiles based on Socio-demographic Index (SDI) levels: low, low middle, middle, high middle, and high SDI levels.

1.3. Method S3:Outcome Measures

Primary outcome metrics extracted for this study were the numbers of mortality of IHD due to kidney dysfunction and the numbers of age-standardised mortality rate (ASMR) associated with IHD related to kidney dysfunction.

To capture uncertainty of the estimate, 95% uncertainty intervals (UIs) were available for the estimates reported.

1.4. Method S4:Statistical Analysis of Temporal Trends

For estimation of magnitude of change of mortality over the study period, a Percentage Change in Age-Standardised Rates (PC-ASR) was calculated. PC-ASR summarises the relative variation in mortality levels at the beginning and end points of the given interval and was calculated using the following formula:

$$\mathrm{PC-ASR} = \left(\frac{\mathrm{ASMR}_{2021} - \mathrm{ASMR}_{1990}}{\mathrm{ASMR}_{1990}}\right) \times 100\%$$

A positive PC-ASR value indicates overall increase in mortality and a negative value indicates overall decrease in mortality [3, 4].

Similarly, Estimated Annual Percentage Change (EAPC) was used to estimate annual trends in age-standardised rates (ASRs). EAPC assumes a linear change in the natural logarithm of ASRs with time, using the regression model $Y = \alpha + \beta X + \varepsilon$, where Y is the natural logarithm of ASR (*ln(ASRs)*), X is the calendar year, α is the constant term, β is the slope term, and ε is the error term. The value of EAPC was calculated from the slope term (β) by the following formula:

$$\mathrm{EAPC} = 100 \times (e^{\beta} - 1)$$

From the linear model, we also calculated the 95% CIs for EAPC. An increasing trend of incidence or mortality is identified if EAPC and its lower 95% CI are positive, and a decreasing trend if the EAPC and its upper 95% CI are negative [3, 4].

1.5. Method S5:Decomposition Analysis

Decomposition of drivers of temporal trends in CKD-attributable IHD deaths. To quantify the drivers of temporal trend in CKD-attributable IHD mortality, a decomposition analysis was performed using the decomposition method previously published by Das Gupta [5, 6]. This decomposition method partitions the total change in deaths into three components: population growth, population ageing (change in population age structure), and epidemiological change (change in the age-specific death rate).

Counterfactuals were generated to estimate the contribution of each component. In the first counterfactual (scenario 1), total population numbers were set to 2023 levels, and population age structure and age-specific death rates were held constant to the 1994 levels. In the second counterfactual (scenario 2), both population size and population age structure were set to the 2023 levels, but age-specific death rates were held constant at 1994 base levels. Contribution of population growth was estimated by subtraction of the observed deaths in 1994 from estimates from scenario 1. The contribution of population aging was calculated as a subtraction of estimate from scenario 1 from estimate from scenario 2. Finally, the contribution of the epidemiological changes that reflect changes in age-specific rates was estimated by subtraction of estimate from scenario 2 from the observed deaths in 2023 [5].

1.6. Method S6:Classification of Disease Burden Transition Patterns

Our novel approach of systematically exploring the dynamic heterogeneity of disease burden trends involves a bivariate classification framework that integrates study trajectories (based on statistical trend estimation) and demographic decomposition analysis. This framework disaggregates the transition patterns along two dimensions: the directionality of epidemiological trends and reasons for the trend identified in the decomposition analysis. Dimension 1: Overall trend pattern The first dimension captures the trajectory of disease burden by the degree of directionality for epidemiological trends, defined by the direction of the PC-ASMR and EAPC over the study period (1994–2023). Pattern of change. We classified this dimension into three patterns: (1) sustained decline pattern, which ascribes to a concordance in both PC-ASMR and EAPC value below zero (<0), defining a relatively good pattern of mortality risk reduction; (2) sustained increase pattern, which ascribes to both values above zero (>0) which represents a standard pattern of upward trend; and (3) mixed effect pattern: for the concordance of PC-ASMR and EAPC with the sign in opposite, indicating an epidemiological phenomenon of nature or change fluctuation. Dimension 2: Driving mechanism. The second dimension characterises the net effect from demographic change and epidemiological trend. The net effect was determined by the summing of the contributions from age structure effect (a effect), population size effect (p effect) and mortality effect

(r effect). For the A Positive net effect pattern, the overall sum of these components was > 0, meaning that driving forces as a whole were promoting an increase in absolute burden of disease. For the A Negative net effect pattern, the overall sum of these components was < 0, meaning that driving forces as a whole were suppressing overall burden. The integration of Transition Patterns Cross-tabulation of these two dimensions generated a 5-category typology to describe burden transition: Category 1 (Sustained decline–negative net effect): States of a decline of mortality where the reductive action of epidemiological improvement overcame the broader expansionary pressures of population size and ageing. Category 2 (Sustained decline–positive net effect): States of a decline of mortality too limited to cancel out the burden expansion driven by population size and ageing. Category 3 (Divergent trend pattern): Refers to areas with the alternating directional signals for absolute rate change and yearly trend estimate. Category 4 (Sustained increase-net effect negative): Unchanged mortality increase but net-negative impact of absolute burden due to substantial offsetting from population data. Category 5 (Sustained increase-net effect positive): Death rates and population show concurrent increase, resulting in double increase of disease burden.

1.7. Method S7:Age-Period-Cohort Analysis

Retaining the temporal trend of CKD-attributable IHD burden in its demographic components, we use the age-period-cohort (APC) model approach. This enables a comprehensive evaluation of the temporal trend, providing insight into the independent effect of chronological age, calendar period and birth cohort [7]. Data on CKD-attributable IHD burden and associated demographic structures were extracted from the GBD 2023 database for years 1994–2023 for individuals aged 25 years and older to 95+ years. Data stratification and parameterization The analysis was stratified into consecutive 5 year age groups from 25–29 years age groups to 95+ years age group. The temporal domain (1994–2023) was segmented into six 5 year calendar periods (1994–1998 to 2019–2023). In turn, the analysis used partially overlapping birth cohort frames that were spaced out by 5 year interval periods (1899–1999). Changes in the overall annual percentage increase in incidence, or net drift, over the full study period (that is, incorporating the

effects of calendar time and cohort succession), and the local drift (or local annual percentage change) by age, which represents age heterogeneity in temporal trend. In the APC model, four primary model parameters characterized secular trends: Period and cohort rate ratios (RR): the relative risk of a specific period or birth cohort relative to the reference period or birth cohort chosen (in this case, the 2009-2013 calendar period, or the 1949 birth cohort); net drift: representing the overall annual percentage change in incidence over the entire study period taking account of calendar time and cohort succession. The significance of trends and model parameters was evaluated by the Wald $\chi 2$ test with $P < 0.05$ considered as significance threshold.

1.8. Method S8:Correlation Analysis with Socio-demographic Index

Finally, to investigate whether development level between regions is associated with burden of disease, we used Spearman's rank correlation test to study the relationship between SDI and ASMR at a national level. To account for multiple hypothesis testing, we used the Benjamini-Hochberg procedure to control for the False Discovery Rate (FDR). We set statistical significance as a adjusted P-value < 0. We also aimed to better address the complex non-linear trends within this relationship using locally weighted scatterplot smoothing (LOWESS) regression to fit the dynamic trend of ASMR along the entire spectrum of sociodemographic development [8].

1.9. Method S9:Frontier Analysis

To capture the achievable minimum disease burden in relation to sociodemographic development, we did a stochastic frontier analysis using the SDI and ASMR measures. This analysis defined a theoretical"frontier"by the minimum ASMR ever achieved at any particular level of the SDI. The magnitude of the gap between the observed ASMR and this theoretical frontier ("effective difference") was computed to infer the possible modification capacity. Frontier curve was estimated by using a data envelop approach, by local weighted scatterplot smoothing (local linear smoothing: LOESS) and local polynomials. The non-parametric approach led to flexible modelling of the non-linear dependence

structure of development status and mortality risk. To capture sufficient stability for estimation and to consider the uncertainty in estimation, 1000 bootstrap resamplings were applied. The final frontier was then the mean ASMR obtained in these resampling iterations, setting a sound benchmark for performance assessment [8].

1.10. Method S10:Data Sources for Mendelian

Randomization Summary statistics for genome-wide association studies (GWAS) of both exposure and outcome variables were obtained from several large repositories of GWAS data. These included US veterans, Finngen consortium, GWAS Catalog, UK Biobank (UKB) databases. Information on number of samples, sample characteristics for each data set can be found in Supplementary Tables. For a robust genotype imputation and estimation of linkage disequilibrium (LD), the genotyping reference panel used was the 1000 Genomes Project (1000G) (http://www.internationalgenome.org), which is a multi-ethnic reference panel set consisting of people of European, Asian and African descent and was used to impute untyped variants [9]. By doing so, the genomic coverage and statistical power of the GWAS analyses in our study were considerably improved. The reference data used are focused on common variants (minor allele frequency (MAF) ≥ 5%) genotypes [9].

1.11. Method S11:Two-Sample Mendelian Randomization Design

Two-sample mendelian randomization (MR) analyses were performed for the estimates of bidirectional causal effects of CKD and IHD on each other and causal effects of Behavioural, dietary, metabolic, and socioeconomic exposures (obesity, dyslipidaemia, hypertension, smoking, diet, and social determinants of health) on CKD and IHD, respectively (Please see **Table S1**, **Table S2** for the relevant details.). All statistics analyses are performed using the Two-Sample MR package of R software (version 4. 4. 3). Instrumental variable selection and harmonization To fulfil the aims of causal inference, genetic IVs were selected following three basic principles: (1) the IVs should be strongly associated with the exposure; (2) the IVs should be independent of any confounders; and (3) the IVs should affect the outcome exclusively

through the exposure, without horizontal pleiotropy. We identified SNPs associated with each trait at a genome-wide significant threshold ($p < 5 \times 10^{-8}$) in Europe samples. For analyses of African and Asian populations, we relaxed the significant threshold to ($p < 5 \times 10^{-6}$) for selection of more instruments (while not substantially reducing the power). To select independent instruments, we also performed a stringent clumping procedure with LD threshold of ($r^2 < 0.001$) within a 10,000 kb window. Harmonization between the exposure and outcome datasets was done by discarding palindromic SNPs with intermediate allele frequency to avoid ambiguity on the strand. Strength of each instrumental variable (IV) was measured by an F-statistic, which is defined as $F = \beta^2/SE^2$, where β is the estimated SNP-exposure effect, and *SE* is the standard error. We discarded instruments with F-statistic < 10 to limit potential biases caused by weak instrument bias [10–13].

1.12. Method S12:Statistical Analysis and Sensitivity Assessment

IVW method was mainly used as the primary method to assess the causal effect of SNPs under the assumption of balanced pleiotropy. Other complementary methods (MR-Egger method, weighted median method, simple mode and weighted mode) were used to assess the robustness of estimates. Benjamini–Hochberg method was used for correcting for multiple tests and P was reported as an FDR <0. 05. Estimates were presented as odds ratio (OR) with 95% confidence intervals (CI). Sensitivity analyses were used to investigate heterogeneity and pleiotropy. Cochran's Q test was used to detect heterogeneity (p <0.05), and the MR-Egger intercept test was used to identify the directional horizontal pleiotropy (non-zero intercept). MR-PRESSO method was used to detect outlier SNPs and correct them [10–13].

1.13. Method S13:Criteria for Establishing Causality

MR-assisted causality was inferred if the IVW estimate was significant (p <0. 05), robust to leave-one-out sensitivity analysis, without horizontal pleiotropy, and the risk estimates from IVW meet at least one of the following 3 hierarchical requirements: (1) if no heterogeneity existed (MR-Egger intercept, MR-

PRESSO global test p >0. 05), estimates from different methods (MR-Egger, weighted median, weighted mode) were in the same direction; (2) if heterogeneity exists (MR-Egger intercept or MR-PRESSO global test p <0. 05), then the estimate is corrected by MR-PRESSO; (3) If heterogeneity exists, the MR-Egger and weighted median were all significant, and both directionally consistent, and had the similar trend in the weighted mode based estimate [10–13].

1.14. Method S14:Triangulation of Evidence through Concordant Causal Effects

In order to robustly evaluate causal effects of behavioural, dietary, metabolic and socioeconomic risk factors for CKD and IHD, we combined evidence from three separate MR studies in a triangulation approach. The approach was based on three steps: (1) identifying risk factors that were causally associated with CKD and/or IHD; (2) estimating the causal effect of CKD on IHD; and (3) identifying risk factors that were associated with both CKD and IHD with directionally similar effects on the pathway from CKD to IHD. For instance, a given risk factor (e.g. years of educational attainment) will be termed as a triangulated association if it is associated with reduced risk of both CKD and IHD and CKD was causally positively associated with increased risk of IHD. As horizontal pleiotropy could manifest differently between the different analyses, it was important for us to focus on associations with directional concordance to obtain stronger causal evidence and also to minimise bias due to pleiotropy [28].

1 2. **Supplementary Figures**

2

3 **2.1. Figure S1**

4

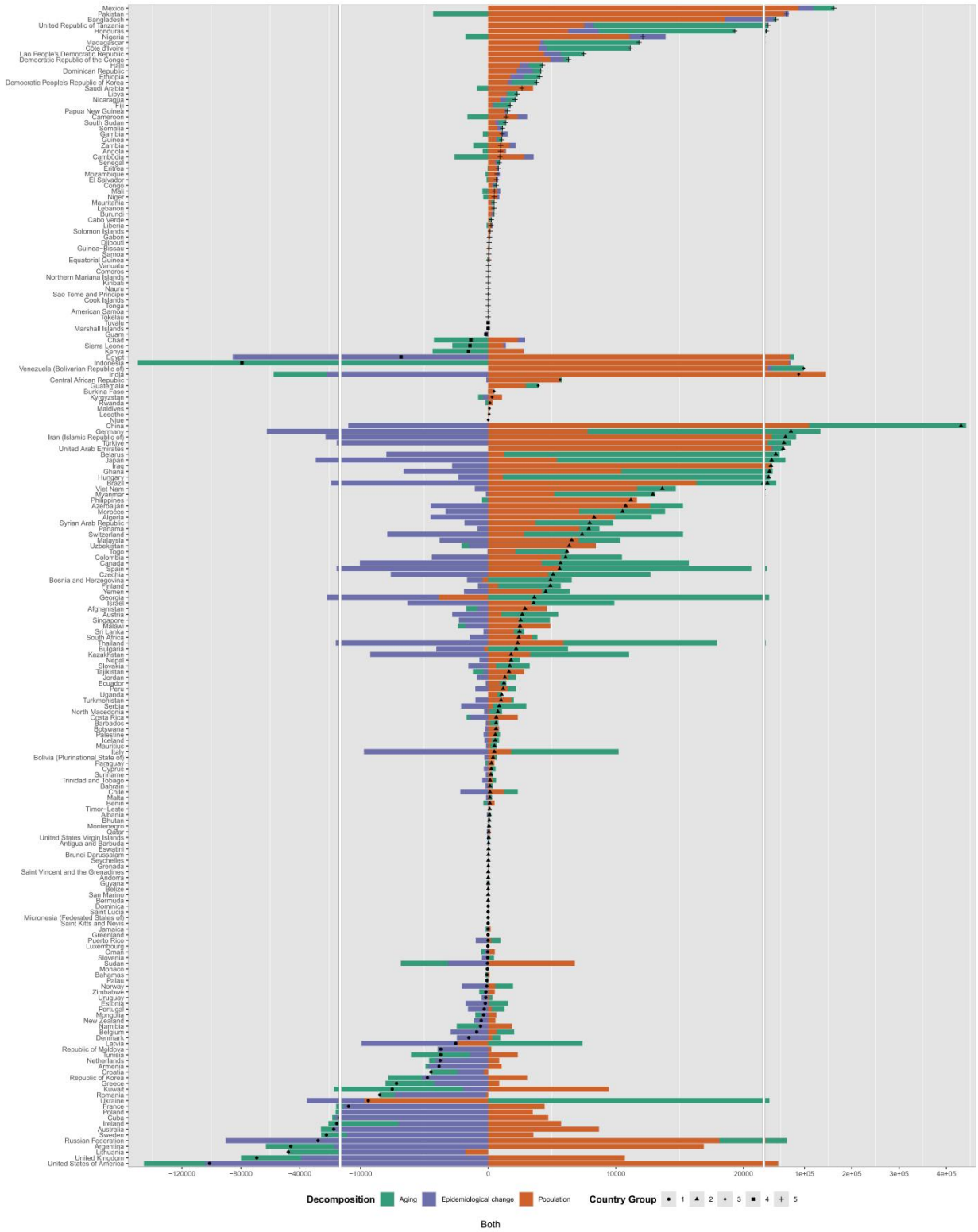
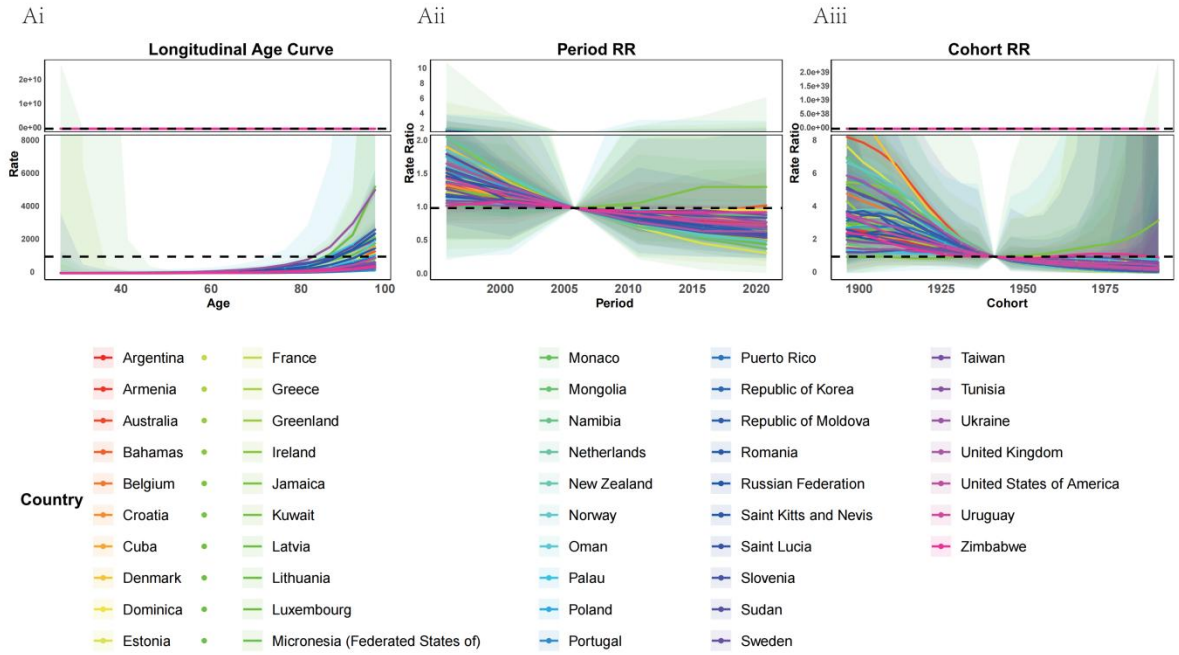


Figure S1. Decomposition in CKD-attributable IHD deaths across 204 countries and territories (1994–2023).

N.B. This figure illustrates the decomposition of absolute differences in ischemic heart disease (IHD) deaths attributable to chronic kidney disease (CKD) over a 30-year period. Total changes in deaths are partitioned into three fundamental drivers utilizing standard demographic decomposition methods: (1) population growth, (2) population aging, and (3) epidemiological change. Countries and territories are stratified into five distinct transition categories (Category 1–5). Within each category, entities are arranged vertically in ascending order of the over difference , marked by specific symbols (●, ▲, ◆, ■ and＋), allowing for a high-resolution visual comparison of the relative magnitudes of demographic versus epidemiological contributions. Positive values signify an escalation in mortality burden, while negative values denote a reduction. More details is explained in the Methods section. Abbreviations: CKD = chronic kidney disease, IHD = ischemic heart disease.

2.2. Figure S2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Lithuania
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Republic of Korea
Republic of Moldova
Romania
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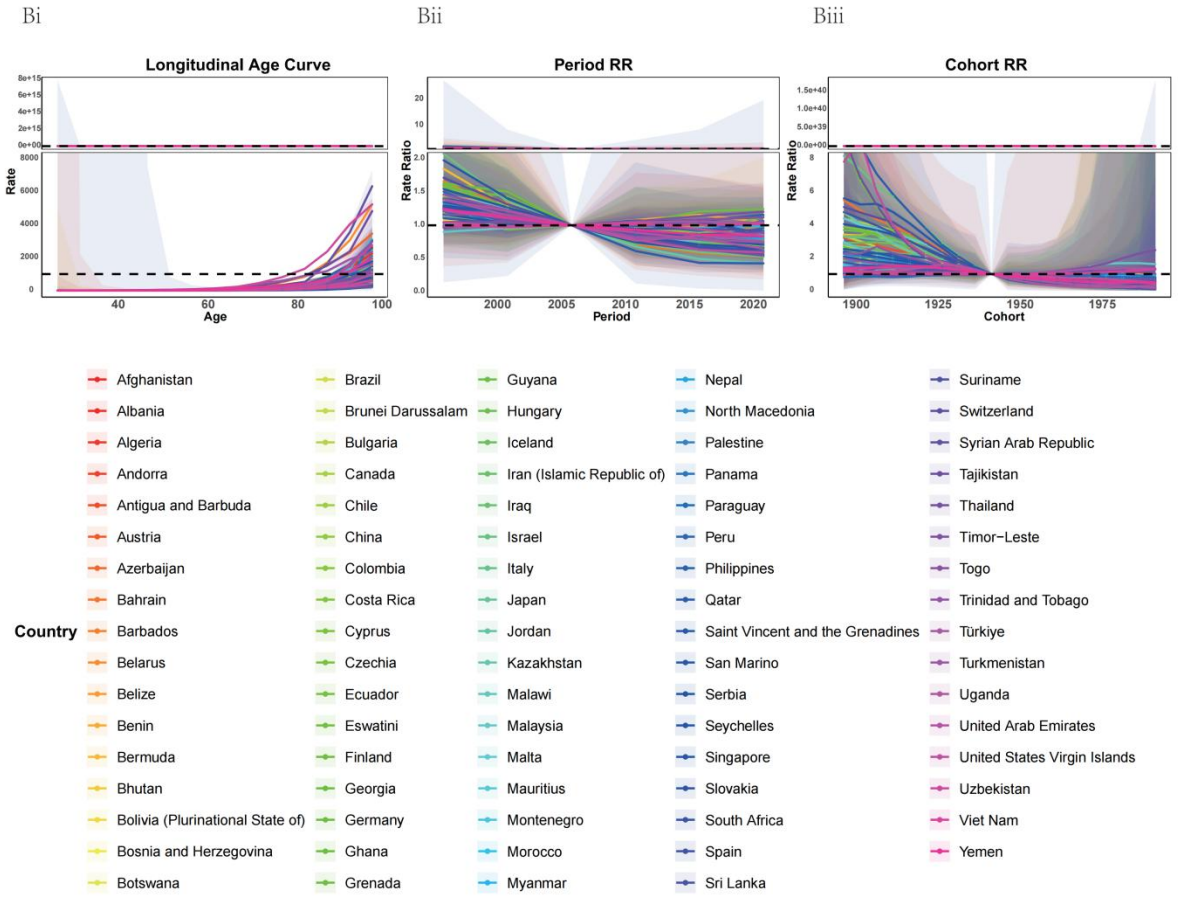
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Bhutan
Bolivia (Plurinational State of)
Bosnia and Herzegovina
Botswana
Brazil
Brunei Darussalam
Bulgaria
Canada
Chile
China
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Costa Rica
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Guyana
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Saint Vincent and the Grenadines
San Marino
Serbia
Seychelles
Singapore
Slovakia
South Africa
Spain
Sri Lanka
Suriname
Switzerland
Syrian Arab Republic
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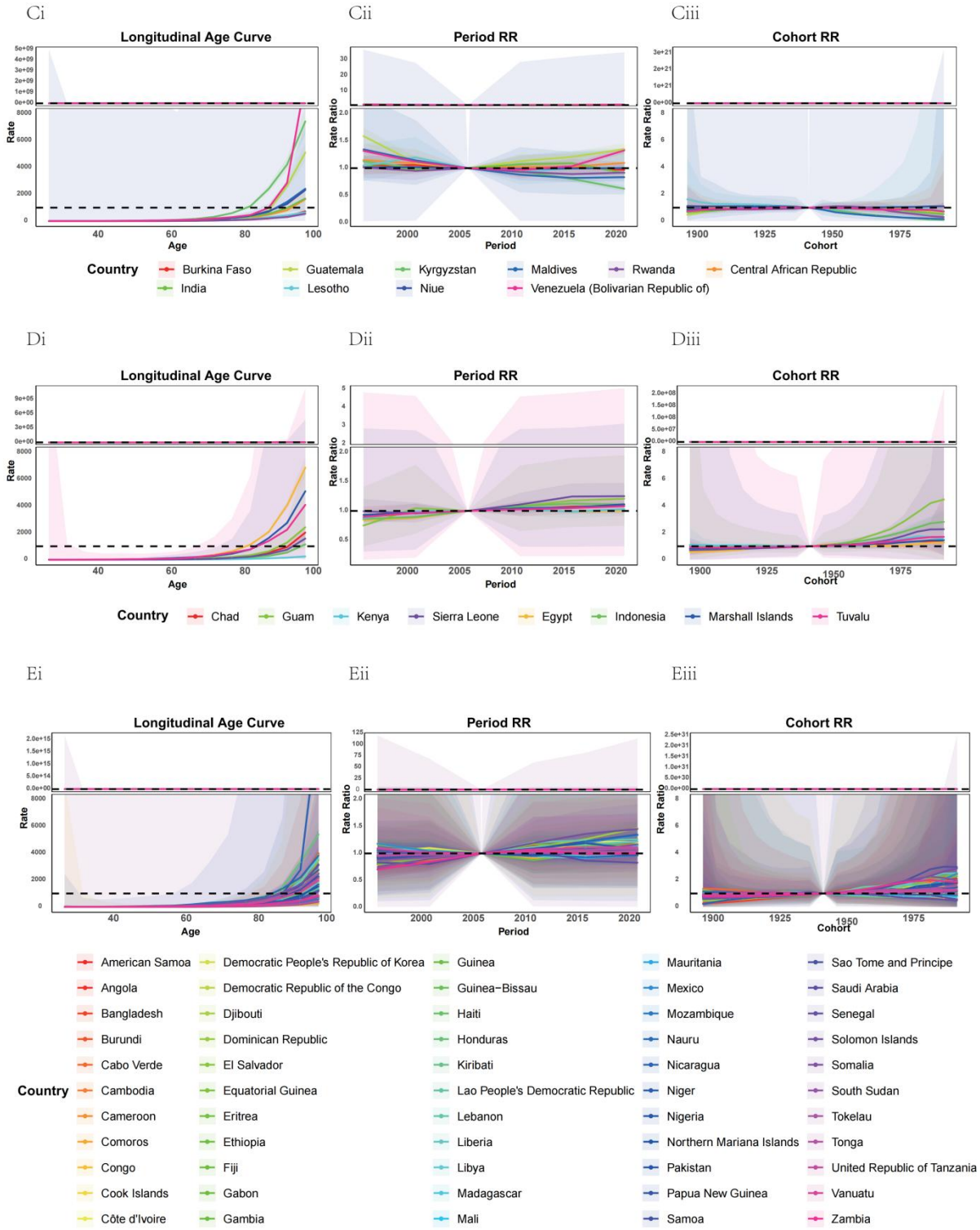
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Bangladesh
Burundi
Cabo Verde
Cambodia
Cameroon
Comoros
Congo
Cook Islands
Côte d'Ivoire
Democratic People's Republic of Korea
Democratic Republic of the Congo
Djibouti
Dominican Republic
El Salvador
Equatorial Guinea
Eritrea
Ethiopia
Fiji
Gabon
Gambia
Guinea
Guinea−Bissau
Haiti
Honduras
Kiribati
Lao People's Democratic Republic
Lebanon
Liberia
Libya
Madagascar
Mali
Mauritania
Mexico
Mozambique
Nauru
Nicaragua
Niger
Nigeria
Northern Mariana Islands
Pakistan
Papua New Guinea
Samoa
Sao Tome and Principe
Saudi Arabia
Senegal
Solomon Islands
Somalia
South Sudan
Tokelau
Tonga
United Republic of Tanzania
Vanuatu
Zambia

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Figure S2. Age–period–cohort analysis of mortality from chronic kidney disease–attributable ischemic heart disease across 204 countries and territories, 1994–2023.

N.B. This ensemble visualizes the longitudinal and cross-sectional dynamics of the CKD-attributable IHD mortality burden utilizing Age–Period–Cohort (APC) modeling. Countries and territories are stratified into five transition categories: A (Category 1), B (Category 2), C (Category 3), D (Category 4), and E (Category 5). For each category, three distinct effects are estimated: (i) Longitudinal age curves (e.g., Ai, Bi), representing the fitted age- specific mortality rates (per 100,000 population) adjusted for period deviations, which reflect the intrinsic biological risk associated with senescence; (ii) Period rate ratios (RRs) (e.g., Aii, Bii), depicting temporal variations in mortality risk across calendar periods relative to the reference, reflecting the impact of evolving healthcare delivery and clinical interventions; and (iii) Cohort rate ratios (RRs) (e.g., Aiii, Biii), characterizing generational shifts in risk and the long-term influence of early-life exposures across successive birth cohorts. Shaded areas represent 95% confidence intervals, and individual colored lines correspond to specific countries within each category. Abbreviations: CKD = chronic kidney disease, IHD = ischemic heart disease, RR = rate ratio.

1 **2.3. Figure S3**

2

Population	Outcome	Trait	nSNP	OR (95% CI)	FDR	significance
African	IHD	Total cholesterol levels	56	0.829 (0.769.C0.894)	<0.001	***
Asian	CKD	Oily fish intake	215	0.616 (0.574.C0.661)	<0.001	***
		Total cholesterol levels	9430	0.685 (0.640.C0.733)	<0.001	***
		Lamb/mutton intake	56	0.745 (0.635.C0.875)	0.001	**
		Smoking status	69	1.103 (1.049.C1.159)	<0.001	***
		Glycated hemoglobin levels	1780	1.529 (1.295.C1.804)	<0.001	***
		Serum creatinine levels	22628	1.531 (1.454.C1.612)	<0.001	***
		Cooked vegetable intake	92	1.536 (1.278.C1.845)	<0.001	***
		Salad / raw vegetable intake	17	2.444 (1.714.C3.485)	<0.001	***
		Waist circumference	294	4.481 (3.760.C5.341)	<0.001	***
	IHD	HDL cholesterol	12758	0.508 (0.493.C0.523)	<0.001	***
		Salad / raw vegetable intake	17	0.678 (0.499.C0.921)	0.028	*
		Average total household income before tax	95	0.692 (0.647.C0.740)	<0.001	***
		Pork intake	27	0.749 (0.627.C0.895)	0.004	**
		Oily fish intake	215	0.768 (0.722.C0.817)	<0.001	***
		Cheese intake	33	0.799 (0.700.C0.912)	0.003	**
		Smoking status	69	0.844 (0.798.C0.892)	<0.001	***
		Serum uric acid levels	25872	1.192 (1.161.C1.224)	<0.001	***
		Cereal intake	42	1.282 (1.081.C1.520)	0.011	*
		Poultry intake	63	1.288 (1.089.C1.523)	0.008	**
		Dried fruit intake	81	1.317 (1.170.C1.482)	<0.001	***
		Waist circumference	294	1.665 (1.460.C1.899)	<0.001	***
		Glycated hemoglobin levels	1772	1.774 (1.471.C2.139)	<0.001	***
		Total cholesterol levels	9417	1.884 (1.784.C1.989)	<0.001	***
		LDL cholesterol	5467	2.242 (2.140.C2.350)	<0.001	***
		Serum creatinine levels	22638	2.997 (2.837.C3.166)	<0.001	***
		Smoking behaviors : Smoking initiation	827	4.175 (2.335.C7.464)	<0.001	***
European	CKD	Cheese intake	59	0.621 (0.518.C0.744)	<0.001	***
		Average total household income before tax	44	0.643 (0.538.C0.769)	<0.001	***
		education	66	0.665 (0.573.C0.773)	<0.001	***
		HDL cholesterol	326	0.843 (0.801.C0.888)	<0.001	***
		phenoage	53	1.030 (1.013.C1.048)	0.003	**
		Glycine levels	50	1.120 (1.047.C1.198)	0.004	**
		triglycerides	281	1.221 (1.156.C1.290)	<0.001	***
		Serum uric acid levels	231	1.230 (1.151.C1.314)	<0.001	***
		smoking initiation	84	1.269 (1.163.C1.385)	<0.001	***
		Townsend deprivation index3	6	1.332 (1.181.C1.501)	<0.001	***
		Glycated hemoglobin levels	68	1.405 (1.148.C1.720)	0.004	**
		Fasting glucose	60	1.426 (1.235.C1.647)	<0.001	***
		Smoking status	114	1.559 (1.299.C1.871)	<0.001	***
		Creatinine levels	331	1.588 (1.517.C1.663)	<0.001	***
		Body mass index (BMI)	417	1.721 (1.642.C1.804)	<0.001	***
		Waist circumference	355	1.914 (1.787.C2.050)	<0.001	***
		Poultry intake	6	2.832 (1.432.C5.601)	0.009	**
	IHD	Average total household income before tax	45	0.649 (0.492.C0.854)	0.010	*
		HDL cholesterol	329	0.784 (0.722.C0.851)	<0.001	***
		triglycerides	290	1.267 (1.172.C1.369)	<0.001	***
		Total cholesterol	56	1.422 (1.236.C1.635)	<0.001	***
		LDL cholesterol	163	1.486 (1.301.C1.698)	<0.001	***
		Body mass index (BMI)	426	1.656 (1.518.C1.805)	<0.001	***
		Smoking status	113	1.665 (1.235.C2.245)	0.007	**
		Waist circumference	352	1.859 (1.651.C2.093)	<0.001	***
		Coffee intake	38	1.925 (1.297.C2.858)	0.010	**

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Figure S3. Multi-ancestry Mendelian randomization estimates for the causal effects of modifiable exposures on chronic kidney disease and ischemic heart disease.

N.B. This forest plot characterizes the causal associations between behavioural, dietary, metabolic, and socioeconomic exposures and the risks of chronic kidney disease (CKD) and ischemic heart disease (IHD) across three ancestral populations (European, East Asian, and African). Causal estimates are presented as odds ratios (ORs) per genetically predicted increase in each exposure, with horizontal bars representing 95% confidence intervals (CIs). The number of instrumental single-nucleotide polymorphisms (nSNP) utilized for each exposure is specified. Statistical significance was determined using the false discovery rate (FDR) to account for multiple testing, categorized as: ***FDR < 0.001; **FDR < 0.01; and *FDR < 0.05. Collectively , these findings delineate a network of metabolic and lifestyle determinants influencing the cardio–renal axis across diverse genetic backgrounds. Abbreviations: CKD, chronic kidney disease; FDR, false discovery rate; IHD, ischemic heart disease; MR, Mendelian randomization; OR, odds ratio.

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